Lab1: Java Fundamentals and Control Structures

Exercise 1: Temperature Converter (Types and Arithmetic)

Write a program that converts a temperature from Celsius to Fahrenheit.

- **Requirements:**
  - Prompt the user to enter a temperature in Celsius as a `double`.
  - Calculate the Fahrenheit equivalent using the formula: ***F = (Celsius * 9/5) + 32***.
  - **Constraint:** Ensure you avoid the "Integer Division Issue" by using `9.0/5` or explicit casting so the decimal part is not truncated.
  - Display the result to the user.

Exercise 2: Leap Year Checker (Conditional Logic)

Create a class called `LeapYear` to determine if a given year is a leap year.

- **Requirements:**
  - Read an integer year from the keyboard using the `Scanner` class.
  - A year is a leap year if it is divisible by 4 but not by 100, OR if it is divisible by 400.
  - Use `if-else` structures to test these conditions.
  - Display "Leap Year" or "Not a Leap Year" accordingly.

Exercise 3: Simple Calculator (Switch Case)

Develop a basic calculator that performs one of four operations (+, -, *, /) based on user input.

- **Requirements:**
  - Ask the user to enter two numbers (`double`) and a character representing the operator.
  - Use a `switch` statement to execute the corresponding arithmetic operation based on the operator entered.
  - Include a `default` case to handle invalid operators.
  - Display the final calculation (e.g., "5.0 + 3.0 = 8.0").

Exercise 4: Multiplication Table (Determinate Loops)

Write a program that displays the multiplication table for a number entered by the user.

- **Requirements:**
  - Prompt the user for an integer N.
  - Use a `for` loop to calculate and display the product of N multiplied by every integer from 1 to 10.

- **Example Output:** If N=5, the program should print: `5 x 1 = 5`, `5 x 2 = 10`, etc.

Exercise 5: Number Guessing Game (Indefinite Loops)

Create a program where the user must guess a predefined "secret number."

- **Requirements:**
  - Initialize a constant `SECRET_NUMBER` (e.g., 42) using the `final` modifier.
  - Use a `do...while` loop to repeatedly ask the user for their guess.
  - The loop should continue as long as the user's guess is incorrect.
  - Inside the loop, provide feedback if the guess is "Too high" or "Too low" using `if-else if` statements.
  - Once the correct number is guessed, display a congratulatory message and the number of attempts.